

Surface Mount Frequency Mixer

ADE-1L+

Level 3 (LO Power +3 dBm) 2 to 500 MHz

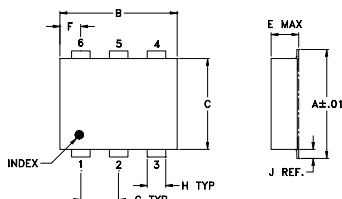
Maximum Ratings

Operating Temperature	-40°C to 85°C
Storage Temperature	-55°C to 100°C
RF Power	50mW
IF Current	40mA
Permanent damage may occur if any of these limits are exceeded.	

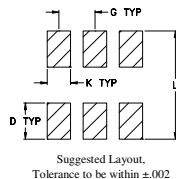
Pin Connections

LO	6
RF	3
IF	2
GROUND	1,4,5

Outline Drawing



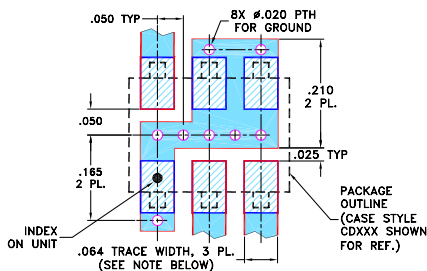
PCB Land Pattern



Outline Dimensions (inch/mm)

A	B	C	D	E	F	G
.272	.310	.220	.100	.112	.055	.100
6.91	7.87	5.59	2.54	2.84	1.40	2.54
H	J	K	L		wt	
.030	.026	.065	.300		grams	
0.76	0.66	1.65	7.62		0.20	

Demo Board MCL P/N: TB-03 Suggested PCB Layout (PL-052)



- NOTES: 1. TRACE WIDTH IS SHOWN FOR ROGERS RO4350B WITH DIELECTRIC THICKNESS .030" ± .002"; COPPER: 1/2 OZ. EACH SIDE. FOR OTHER MATERIALS TRACE WIDTH MAY NEED TO BE MODIFIED.
2. BOTTOM SIDE OF THE PCB IS CONTINUOUS GROUND PLANE.
- DENOTES PCB COPPER LAYOUT WITH SMOBC (SOLDER MASK OVER BARE COPPER)
■ DENOTES COPPER LAND PATTERN FREE OF SOLDER MASK

Features

- low conversion loss, 5.2 dB typ.
- excellent L-R isolation, 55 dB typ.
- low profile package
- aqueous washable
- protected by US patent 6,133,525

Applications

- cellular
- instrumentation



Generic photo used for illustration purposes only
CASE STYLE: CD542

+RoHS Compliant

The +Suffix identifies RoHS Compliance. See our web site for RoHS Compliance methodologies and qualifications



Available Tape and Reel at no extra cost

Reel Size	Devices/Reel
7"	20, 50, 100, 200
13"	500, 1000

Electrical Specifications

FREQUENCY (MHz)		CONVERSION LOSS (dB)				LO-RF ISOLATION (dB)			LO-IF ISOLATION (dB)			IP3 at center band (dBm)						
LO/RF	IF	Mid-Band m		Total Range		L	M	U	L	M	U							
		$\bar{X}$	σ	Max.	Max.													
		Typ. Min.																
2-500	DC-500	5.2	0.10	7.2	8.0	68	50	55	30	44	30	55	40	45	30	35	25	16

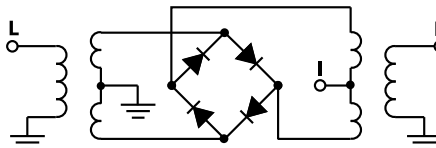
1 dB COMP: 0 dBm typ.

L = low range [f_L to $10 f_L$] M = mid range [$10 f_L$ to $f_U/2$] U = upper range [$f_U/2$ to f_U]
m = mid band [$2f_L$ to $f_U/2$]

Typical Performance Data

Frequency (MHz)		Conversion Loss (dB)	Isolation L-R (dB)	Isolation L-I (dB)	VSWR RF Port (:1)	VSWR LO Port (:1)
RF	LO					
		LO +3dBm	LO +3dBm	LO +3dBm	LO +3dBm	LO +3dBm
2.00	32.00	5.88	68.80	57.20	1.32	1.67
5.00	35.00	5.44	68.30	56.40	1.19	1.66
10.00	40.00	5.32	68.00	55.40	1.14	1.66
20.00	50.00	5.25	67.40	54.20	1.10	1.80
50.00	80.00	5.34	63.80	49.80	1.08	1.84
75.00	105.00	5.30	60.80	47.40	1.07	1.84
80.09	110.09	5.32	60.50	46.90	1.07	1.85
100.00	130.00	5.32	59.20	45.40	1.06	1.85
150.00	180.00	5.32	56.30	43.20	1.05	1.96
160.09	190.09	5.31	50.60	42.90	1.10	1.67
200.00	230.00	5.34	48.80	41.60	1.10	1.69
240.08	270.08	5.37	47.40	40.30	1.10	1.73
250.00	280.00	5.38	47.20	40.00	1.10	1.73
300.00	330.00	5.52	44.30	39.10	1.11	1.80
320.07	350.07	5.50	43.30	38.60	1.12	1.82
400.00	430.00	5.68	41.80	35.30	1.19	2.01
450.00	480.00	5.73	41.50	34.20	1.23	2.04
470.00	500.00	5.69	41.50	33.70	1.24	2.16
480.00	510.06	5.67	40.90	33.40	1.24	2.18
500.00	530.00	5.69	40.70	33.00	1.25	2.25

Electrical Schematic



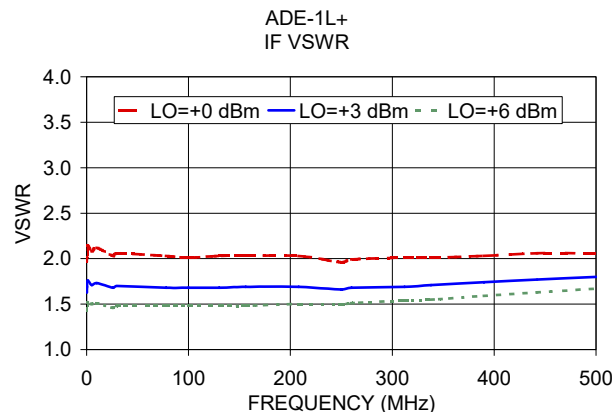
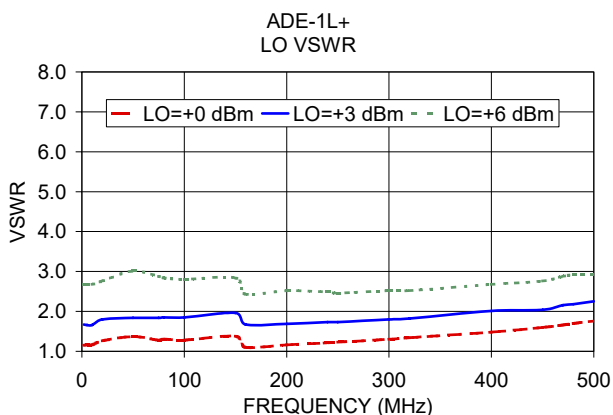
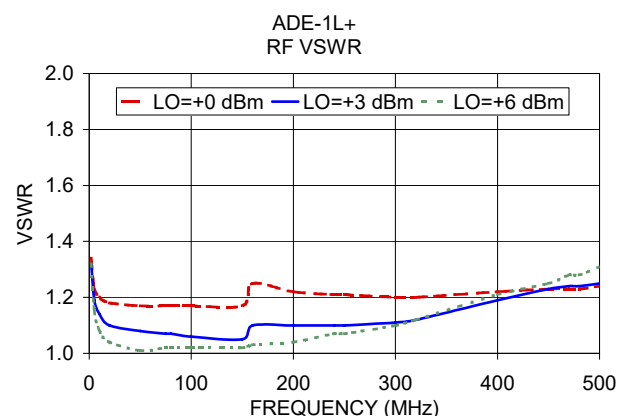
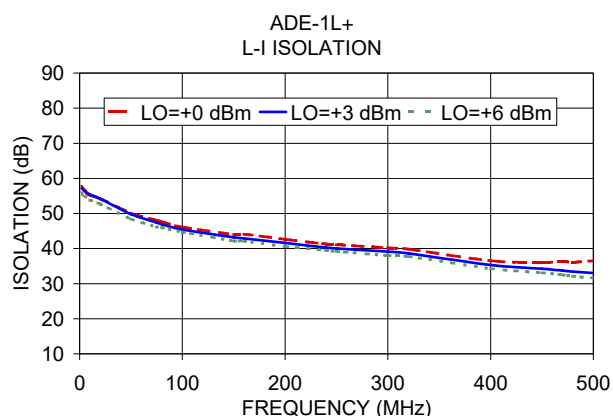
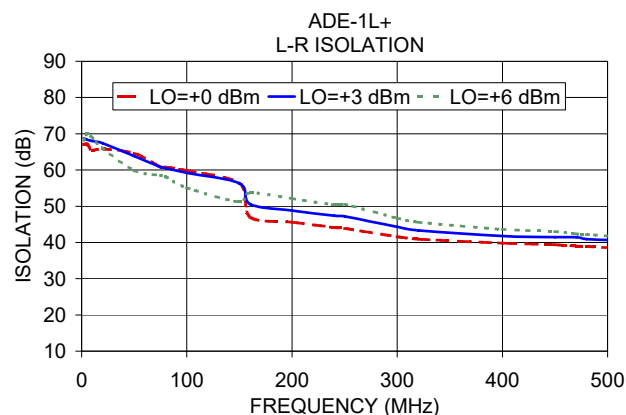
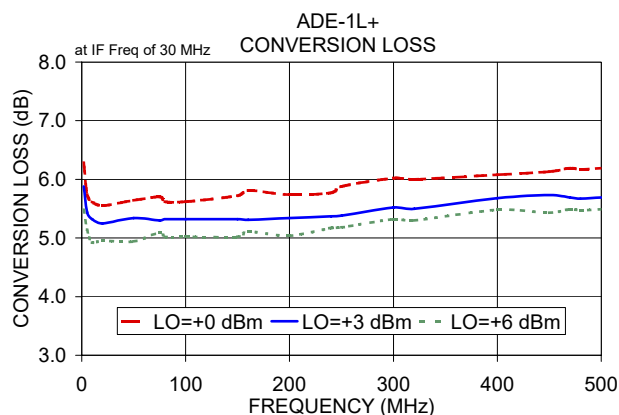
Notes

- A. Performance and quality attributes and conditions not expressly stated in this specification document are intended to be excluded and do not form a part of this specification document.
B. Electrical specifications and performance data contained in this specification document are based on Mini-Circuit's applicable established test performance criteria and measurement instructions.
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Frequency Mixer

ADE-1L+

Typical Performance Data

RF (IN) (MHz)	LO (MHz)	CONVERSION LOSS IF FIXED @IF(OUT)=30MHz (dB)		
		@LO (dBm)		
		0	+3	+6
2.0	32.0	6.28	5.88	5.48
5.0	35.0	5.74	5.44	5.14
10.0	40.0	5.62	5.32	4.92
50.4	80.4	6.82	6.00	5.62
90.8	120.8	6.82	6.02	5.64
110.9	140.9	6.79	6.00	5.62
151.3	181.3	6.76	5.99	5.65
171.5	201.5	6.70	5.98	5.66
211.8	241.8	6.68	6.00	5.68
232.0	262.0	6.66	6.01	5.71
272.3	302.3	6.69	6.06	5.73
292.5	322.5	6.69	6.04	5.73
332.8	362.8	6.74	6.09	5.78
353.0	383.0	6.76	6.15	5.81
393.3	423.3	6.80	6.21	5.84
453.8	483.8	6.88	6.29	5.97
474.0	504.0	6.91	6.27	5.98
514.3	544.3	7.02	6.32	5.98
534.5	564.5	7.07	6.36	6.00
574.8	604.8	7.32	6.54	6.04
595.0	625.0	7.46	6.69	6.13
635.4	665.4	7.77	7.07	6.44
655.5	685.5	7.91	7.22	6.58
695.9	725.9	8.26	7.54	6.93
756.4	786.4	8.70	7.94	7.36
776.5	806.5	8.87	8.11	7.53
816.9	846.9	8.99	8.26	7.69
837.0	867.0	9.06	8.31	7.75
877.4	907.4	9.14	8.36	7.79
897.6	927.6	9.11	8.31	7.73
937.9	967.9	9.24	8.35	7.73
958.1	988.1	9.39	8.39	7.75
998.4	1028.4	9.47	8.39	7.76
1018.6	1048.6	9.59	8.42	7.84
1058.9	1088.9	9.88	8.63	8.08
1079.1	1109.1	9.90	8.72	8.23
1119.4	1149.4	10.34	9.10	8.61
1139.6	1169.6	10.79	9.37	8.85
1179.9	1209.9	11.41	9.84	9.34
1200.1	1230.1	11.67	10.07	9.56

RF (IN) (MHz)	LO (MHz)	IP3 INPUT (dBm)		
		@LO (dBm)		
		0	+3	+6
10.1	40.1	25.46	20.71	26.68
50.4	80.4	20.07	21.62	18.40
90.8	120.8	24.82	21.06	20.06
110.9	140.9	21.44	21.23	27.99
151.3	181.3	30.54	24.17	26.06
171.5	201.5	21.78	18.56	20.59
211.8	241.8	21.01	23.19	20.97
232.0	262.0	18.26	17.17	18.99
272.3	302.3	37.25	19.10	23.66
292.5	322.5	19.48	15.80	14.82
332.8	362.8	29.38	15.64	14.61
353.0	383.0	18.46	21.54	15.73
393.3	423.3	19.56	16.83	15.87
413.5	443.5	21.99	16.37	16.25
453.8	483.8	16.11	13.51	12.69
474.0	504.0	17.91	13.10	11.57
514.3	544.3	13.20	13.67	12.57
534.5	564.5	11.42	13.27	13.43
574.8	604.8	7.92	11.15	14.18
595.0	625.0	6.64	9.15	14.05
635.4	665.4	4.50	5.62	9.50
655.5	685.5	3.79	4.48	7.19
695.9	725.9	3.68	4.08	5.64
716.0	746.0	3.95	4.44	5.70
756.4	786.4	4.54	5.19	6.33
776.5	806.5	4.95	5.48	6.61
816.9	846.9	6.17	6.79	8.02
837.0	867.0	6.88	8.12	9.59
877.4	907.4	8.77	9.97	12.72
897.6	927.6	10.17	10.59	14.01
937.9	967.9	12.01	12.44	14.89
958.1	988.1	11.93	14.49	17.78
998.4	1028.4	12.16	13.51	16.27
1018.6	1048.6	11.73	13.27	14.23
1058.9	1088.9	10.95	12.09	13.62
1079.1	1109.1	10.58	11.55	12.99
1119.4	1149.4	9.65	11.34	11.13
1139.6	1169.6	8.95	10.07	10.79
1179.9	1209.9	8.48	9.53	10.22
1200.1	1230.1	8.51	9.17	11.05

RF (IN) (MHz)	LO (MHz)	COMPRESSION @RF IN=0dBm (dB)		
		@LO (dBm)		
		0	+3	+6
10.1	40.1	1.34	0.82	0.56
50.4	80.4	1.35	0.78	0.48
90.8	120.8	1.34	0.78	0.51
110.9	140.9	1.35	0.81	0.51
151.3	181.3	1.29	0.78	0.48
171.5	201.5	1.29	0.78	0.47
211.8	241.8	1.27	0.70	0.44
232.0	262.0	1.31	0.73	0.45
272.3	302.3	1.26	0.72	0.47
292.5	322.5	1.23	0.67	0.46
332.8	362.8	1.23	0.65	0.42
353.0	383.0	1.21	0.66	0.43
393.3	423.3	1.21	0.65	0.46
413.5	443.5	1.27	0.69	0.46
453.8	483.8	1.33	0.75	0.50
474.0	504.0	1.39	0.81	0.51
514.3	544.3	1.39	0.89	0.54
534.5	564.5	1.42	0.92	0.58
574.8	604.8	1.49	1.08	0.70
595.0	625.0	1.51	1.08	0.77
635.4	665.4	1.55	1.17	0.97
655.5	685.5	1.55	1.12	0.98
695.9	725.9	1.42	1.04	0.84
716.0	746.0	1.36	0.99	0.83
756.4	786.4	1.22	0.92	0.74
776.5	806.5	1.20	0.86	0.66
816.9	846.9	1.28	0.89	0.72
837.0	867.0	1.22	0.85	0.69
877.4	907.4	1.32	0.92	0.73
897.6	927.6	1.34	0.96	0.76
937.9	967.9	1.39	1.07	0.81
958.1	988.1	1.45	1.11	0.80
998.4	1028.4	1.50	1.20	0.85
1018.6	1048.6	1.45	1.14	0.83
1058.9	1088.9	1.53	1.11	0.76
1079.1	1109.1	1.51	1.08	0.68
1119.4	1149.4	1.42	1.00	0.64
1139.6	1169.6	1.43	1.08	0.65
1179.9	1209.9	1.37	1.00	0.62
1200.1	1230.1	1.28	0.97	0.57

REV. X2

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IF/RF MICROWAVE COMPONENTS • ISO 9001 ISO 14001 AS 9100 CERTIFIED RoHS compliant

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Typical Performance Data

IF (OUT) (MHz)	LO (MHz)	CONVERSION LOSS VS. IF FREQUENCY @RF(IN)=250.1MHz (dB)	IF (OUT) (MHz)	LO (MHz)	CONVERSION LOSS VS. IF FREQUENCY @RF(IN)=10.1MHz (dB)	IF (OUT) (MHz)	LO (MHz)	CONVERSION LOSS VS. IF FREQUENCY @RF(IN)=500.1MHz (dB)
		@LO (dBm)			@LO (dBm)			@LO (dBm)
		+3			+3			+3
240.0	10.1	5.99	20.2	10.1	5.79	490.0	10.1	6.38
234.1	16.0	6.00	32.2	22.1	5.55	478.0	22.1	6.36
228.2	21.9	5.99	44.2	34.1	5.53	466.0	34.1	6.33
222.3	27.8	5.98	56.2	46.1	5.48	454.0	46.1	6.32
216.4	33.7	5.97	68.2	58.1	5.47	442.0	58.1	6.30
210.5	39.6	5.94	80.2	70.1	5.49	430.0	70.1	6.32
204.6	45.5	5.94	92.2	82.1	5.48	418.0	82.1	6.26
198.7	51.4	5.91	104.2	94.1	5.52	406.0	94.1	6.18
192.8	57.3	5.90	116.2	106.1	5.49	394.0	106.1	6.14
186.9	63.2	5.89	128.2	118.1	5.56	382.0	118.1	6.13
181.0	69.1	5.88	140.2	130.1	5.54	370.0	130.1	6.09
175.1	75.0	5.89	152.2	142.1	5.51	358.0	142.1	6.09
169.2	80.9	5.85	164.2	154.1	5.54	346.0	154.1	6.08
163.3	86.8	5.85	176.2	166.1	5.53	334.0	166.1	6.07
157.4	92.7	5.86	188.2	178.1	5.56	322.0	178.1	6.05
151.5	98.6	5.84	200.2	190.1	5.54	310.0	190.1	6.04
145.6	104.5	5.83	212.2	202.1	5.56	298.0	202.1	6.03
139.7	110.4	5.82	224.2	214.1	5.60	286.0	214.1	6.01
133.8	116.3	5.81	236.2	226.1	5.55	274.0	226.1	6.00
127.9	122.2	5.80	248.2	238.1	5.61	262.0	238.1	6.02
122.1	128.0	5.79	260.2	250.1	5.63	250.0	250.1	5.62
116.2	133.9	5.77	272.2	262.1	5.62	238.0	262.1	5.99
110.3	139.8	5.76	284.2	274.1	5.62	226.0	274.1	5.98
104.4	145.7	5.75	296.2	286.1	5.68	214.0	286.1	6.06
98.5	151.6	5.78	308.2	298.1	5.68	202.0	298.1	6.07
92.6	157.5	5.78	320.2	310.1	5.70	190.0	310.1	6.09
86.7	163.4	5.77	332.2	322.1	5.70	178.0	322.1	6.10
80.8	169.3	5.78	344.2	334.1	5.74	166.0	334.1	6.12
74.9	175.2	5.74	356.2	346.1	5.73	154.0	346.1	6.14
69.0	181.1	5.76	368.2	358.1	5.73	142.0	358.1	6.18
63.1	187.0	5.77	380.2	370.1	5.71	130.0	370.1	6.17
57.2	192.9	5.73	392.2	382.1	5.72	118.0	382.1	6.20
51.3	198.8	5.78	404.2	394.1	5.80	106.0	394.1	6.19
45.4	204.7	5.77	416.2	406.1	6.16	94.0	406.1	6.17
39.5	210.6	5.78	428.2	418.1	6.24	82.0	418.1	6.18
33.6	216.5	5.78	440.2	430.1	5.94	70.0	430.1	6.27
27.7	222.4	5.81	452.2	442.1	5.92	58.0	442.1	6.25
21.8	228.3	5.79	464.2	454.1	5.89	46.0	454.1	6.23
15.9	234.2	5.87	488.2	478.1	5.91	22.0	478.1	6.36
10.0	240.1	6.09	500.2	490.1	6.01	10.0	490.1	6.28

Typical Performance Data

LO (MHz)	LO-RF ISOLATION (dB)			LO-IF ISOLATION (dB)		
	@LO (dBm)			@LO (dBm)		
	0	+3	+6	0	+3	+6
2.0	67.0	68.8	68.2	57.6	57.2	55.8
5.0	67.1	68.3	70.0	56.4	56.4	54.9
10.0	65.5	68.0	68.9	55.2	55.4	53.8
50.4	59.47	63.85	68.78	58.55	56.83	55.58
90.8	54.77	58.66	63.27	53.39	51.80	50.80
110.9	53.17	56.70	60.67	51.53	50.16	49.34
151.3	49.81	53.15	57.34	48.78	47.54	46.76
171.5	48.94	52.21	56.31	47.62	46.56	45.80
211.8	47.23	50.27	54.06	45.78	44.89	44.37
232.0	46.10	49.19	52.80	44.96	44.26	43.62
272.3	44.29	47.35	50.62	43.60	42.93	42.21
292.5	43.26	46.27	49.37	43.40	42.87	42.08
332.8	41.98	44.50	47.25	42.40	41.81	40.79
353.0	41.17	43.44	45.83	42.12	41.45	40.27
393.3	40.25	42.15	43.94	41.13	40.42	39.68
453.8	39.00	40.89	42.37	40.34	38.05	36.46
474.0	38.72	40.40	41.61	40.78	37.97	35.92
514.3	38.45	40.10	41.01	42.12	38.10	35.29
534.5	38.05	39.76	40.61	42.91	38.40	35.16
574.8	36.80	38.74	39.74	42.09	38.52	34.74
595.0	36.17	38.19	39.26	40.62	37.98	34.26
635.4	34.59	36.52	37.64	36.75	35.17	33.03
655.5	34.34	36.18	37.22	35.41	33.69	32.10
695.9	32.95	34.65	35.64	33.48	31.35	29.93
756.4	30.89	32.40	33.42	31.87	29.61	27.84
776.5	30.46	31.96	32.96	31.40	29.11	27.25
816.9	29.60	31.15	32.21	30.15	28.19	26.34
837.0	29.42	30.90	31.89	29.52	27.74	25.91
877.4	28.96	30.36	31.28	28.43	27.20	25.44
897.6	28.46	29.90	30.84	27.62	26.82	25.14
937.9	28.17	29.42	30.10	26.12	26.10	24.48
958.1	27.81	28.96	29.46	25.18	25.53	23.92
998.4	27.89	28.78	28.86	23.27	24.32	22.85
1018.6	27.66	28.27	28.06	22.33	23.46	22.03
1058.9	27.75	27.77	27.09	20.60	22.05	21.02
1079.1	27.63	27.46	26.58	19.50	21.13	20.54
1119.4	27.63	26.72	25.41	17.82	19.47	19.37
1139.6	27.77	26.50	24.98	17.04	18.72	18.86
1179.9	27.85	25.98	24.14	15.54	17.19	17.83
1200.1	27.54	25.51	23.63	14.93	16.55	17.27

RF (IN) (MHz)	LO (MHz)	RF-IF ISOLATION (dB)		
		@LO (dBm)		
		0	+3	+6
10.1	40.1	45.79	52.75	40.38
50.4	80.4	32.33	32.38	32.57
90.8	120.8	27.80	27.99	28.12
110.9	140.9	26.39	26.63	26.76
151.3	181.3	24.27	24.53	24.70
171.5	201.5	23.52	23.80	24.00
211.8	241.8	22.47	22.81	22.99
232.0	262.0	22.08	22.47	22.68
272.3	302.3	21.63	22.08	22.43
292.5	322.5	21.54	22.01	22.37
332.8	362.8	21.59	22.10	22.41
353.0	383.0	21.74	22.28	22.60
393.3	423.3	22.13	22.90	23.34
413.5	443.5	22.19	23.07	23.61
453.8	483.8	22.21	23.27	23.99
474.0	504.0	21.90	22.93	23.55
514.3	544.3	20.76	21.49	21.87
534.5	564.5	20.14	20.62	20.90
574.8	604.8	18.93	19.07	19.12
595.0	625.0	18.45	18.44	18.39
635.4	665.4	17.67	17.52	17.36
655.5	685.5	17.39	17.22	17.05
695.9	725.9	17.06	16.84	16.68
716.0	746.0	16.99	16.73	16.55
756.4	786.4	16.88	16.60	16.35
776.5	806.5	16.89	16.61	16.36
816.9	846.9	16.59	16.39	16.18
837.0	867.0	16.35	16.20	16.00
877.4	907.4	15.68	15.64	15.46
897.6	927.6	15.25	15.23	15.07
937.9	967.9	14.35	14.40	14.34
958.1	988.1	13.85	13.94	13.96
998.4	1028.4	12.94	13.08	13.24
1018.6	1048.6	12.45	12.63	12.81
1058.9	1088.9	11.63	11.81	11.91
1079.1	1109.1	11.20	11.33	11.35
1119.4	1149.4	10.36	10.35	10.26
1139.6	1169.6	9.94	9.85	9.71
1179.9	1209.9	9.17	8.91	8.64
1200.1	1230.1	8.81	8.47	8.14

Frequency Mixer

ADE-1L+

Typical Performance Data

RF (IN) (MHz)	LO (MHz)	RF VSWR (:1)			LO (MHz)	LO VSWR (:1)			IF (OUT) (MHz)	IF VSWR @LO=500.1MHz (:1)		
		@LO (dBm)				@LO (dBm)				@LO (dBm)		
		0	+3	+6		0	+3	+6		0	+3	+6
2.0	32.0	1.34	1.32	1.32	2.0	1.15	1.67	2.68	0.1	2.01	1.66	1.45
5.0	35.0	1.23	1.19	1.15	5.0	1.16	1.66	2.68	0.2	1.97	1.63	1.43
10.0	40.0	1.20	1.14	1.08	10.0	1.16	1.66	2.68	0.5	2.04	1.69	1.48
50.4	80.4	1.34	1.17	1.06	50.4	1.11	1.42	2.13	1.0	2.14	1.76	1.52
90.8	120.8	1.41	1.23	1.12	90.8	1.11	1.42	2.10	5.0	2.08	1.71	1.50
110.9	140.9	1.37	1.18	1.08	110.9	1.12	1.39	2.03	10.0	2.12	1.73	1.51
151.3	181.3	1.36	1.20	1.11	151.3	1.11	1.42	2.09	22.3	2.06	1.77	1.61
171.5	201.5	1.38	1.21	1.11	171.5	1.11	1.43	2.11	34.5	1.99	1.73	1.51
211.8	241.8	1.28	1.13	1.07	211.8	1.12	1.44	2.09	46.8	1.97	1.67	1.51
232.0	262.0	1.29	1.15	1.10	232.0	1.12	1.47	2.15	59.0	1.92	1.63	1.47
272.3	302.3	1.32	1.18	1.12	272.3	1.15	1.51	2.18	71.3	1.93	1.67	1.49
292.5	322.5	1.30	1.16	1.11	292.5	1.16	1.51	2.15	83.5	1.96	1.68	1.52
332.8	362.8	1.30	1.18	1.14	332.8	1.19	1.55	2.20	95.8	2.02	1.73	1.56
353.0	383.0	1.29	1.17	1.14	353.0	1.21	1.58	2.25	108.0	2.03	1.75	1.56
393.3	423.3	1.30	1.21	1.19	393.3	1.27	1.61	2.23	120.3	2.04	1.76	1.58
413.5	443.5	1.30	1.21	1.19	413.5	1.28	1.66	2.31	132.5	2.02	1.74	1.57
453.8	483.8	1.30	1.22	1.21	453.8	1.34	1.73	2.40	144.8	2.04	1.74	1.58
474.0	504.0	1.31	1.24	1.23	474.0	1.38	1.74	2.38	157.0	2.03	1.74	1.57
514.3	544.3	1.31	1.21	1.21	514.3	1.44	1.80	2.43	181.5	2.06	1.77	1.60
534.5	564.5	1.30	1.21	1.22	534.5	1.47	1.83	2.44	193.8	2.11	1.81	1.64
574.8	604.8	1.36	1.25	1.23	574.8	1.55	1.89	2.45	206.0	2.15	1.84	1.67
595.0	625.0	1.38	1.25	1.22	595.0	1.58	1.95	2.53	218.3	2.17	1.86	1.67
635.4	665.4	1.45	1.33	1.27	635.4	1.65	2.06	2.66	230.5	2.15	1.85	1.67
655.5	685.5	1.51	1.39	1.32	655.5	1.69	2.10	2.71	242.8	2.12	1.82	1.65
695.9	725.9	1.63	1.52	1.44	695.9	1.74	2.15	2.78	255.0	2.10	1.80	1.64
716.0	746.0	1.73	1.62	1.55	716.0	1.77	2.16	2.78	267.3	2.09	1.80	1.63
756.4	786.4	1.99	1.87	1.79	756.4	1.85	2.22	2.83	279.5	2.11	1.81	1.65
776.5	806.5	2.07	1.96	1.88	776.5	1.88	2.25	2.86	291.8	2.12	1.82	1.66
816.9	846.9	2.33	2.22	2.15	816.9	1.96	2.31	2.91	304.0	2.13	1.84	1.67
837.0	867.0	2.45	2.34	2.26	837.0	2.01	2.35	2.95	328.5	2.13	1.85	1.67
877.4	907.4	2.55	2.45	2.37	877.4	2.08	2.37	2.92	340.8	2.13	1.85	1.68
897.6	927.6	2.70	2.60	2.52	897.6	2.12	2.36	2.89	353.0	2.13	1.84	1.68
937.9	967.9	2.83	2.70	2.62	937.9	2.23	2.44	2.92	365.3	2.11	1.84	1.68
958.1	988.1	2.84	2.70	2.62	958.1	2.27	2.44	2.92	377.5	2.12	1.85	1.69
998.4	1028.4	3.05	2.90	2.82	998.4	2.36	2.46	2.90	389.8	2.15	1.88	1.72
1018.6	1048.6	3.12	2.95	2.86	1018.6	2.42	2.52	2.96	402.0	2.19	1.92	1.76
1058.9	1088.9	3.14	2.95	2.87	1058.9	2.57	2.60	3.01	426.5	2.21	1.94	1.77
1079.1	1109.1	3.21	3.03	2.95	1079.1	2.65	2.63	2.99	438.8	2.19	1.92	1.76
1119.4	1149.4	3.26	3.04	2.95	1119.4	2.86	2.82	3.16	451.0	2.17	1.90	1.75
1139.6	1169.6	3.34	3.08	2.97	1139.6	3.00	2.93	3.24	463.3	2.16	1.90	1.76
1179.9	1209.9	3.43	3.10	2.96	1179.9	3.21	3.08	3.31	487.8	2.22	1.96	1.81
1200.1	1230.1	3.38	3.03	2.89	1200.1	3.27	3.16	3.38	500.0	2.18	2.06	2.01

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Harmonics Tables

RF HARMONICS ORDER

	(-dBm)	(-dBc)										
0	-	-	18	33	21	27	21	46	36	47	39	57
1	-	16	0	27	14	33	23	38	39	46	41	61
2	107	73	62	63	62	61	58	73	54	70	64	69
3	111	63	70	69	61	86	57	73	59	79	64	83
4	116	89	95	89	85	78	85	83	84	89	92	102
5	121	94	87	96	84	83	77	89	86	97	84	94
6	131	103	102	92	106	96	81	91	92	105	116	101
7	114	101	101	104	98	96	89	77	86	100	101	102
8	118	100	106	114	113	98	101	99	70	93	91	98
9	117	110	108	108	113	105	97	93	100	68	103	92
10	128	107	100	111	104	97	116	97	101	96	67	90
RF CAL	0	1	2	3	4	5	6	7	8	9	10	

LO HARMONICS ORDER

Test conditions: RF IN: 250.1 MHz; -15.00 dBm.
LO IN: 280.01 MHz; +3.00 dBm
IF OUT: 29.91 MHz; -20.89 dBm

RF HARMONICS ORDER

	(-dBm)	(-dBc)										
0	-	-	27	42	33	40	34	54	52	72	59	70
1	-	17	0	28	14	37	24	41	43	52	52	66
2	98	58	50	58	52	63	51	57	48	72	61	67
3	93	45	47	49	50	53	48	51	52	57	55	64
4	90	68	70	69	79	66	77	65	70	71	66	80
5	93	70	67	67	58	69	54	64	53	77	56	73
6	97	83	88	81	93	96	77	87	78	82	79	81
7	93	79	77	86	73	87	74	75	70	74	69	76
8	96	95	92	89	94	88	103	93	89	92	95	92
9	97	102	89	104	81	100	81	92	89	71	96	89
10	95	103	101	103	107	100	97	90	96	94	83	95
RF CAL	0	1	2	3	4	5	6	7	8	9	10	

LO HARMONICS ORDER

Test conditions: RF IN: 250.1 MHz; -5.00 dBm.
LO IN: 280.01 MHz; +3.00 dBm
IF OUT: 29.91 MHz; -11.01 dBm

- Notes: 1. All Harmonics are in (dBc) relative to IF OUTPUT.
2. + entry denotes harmonics are in (dBc) above IF OUTPUT.
3. RF Cal represent the Harmonics level of the RF input signal to the mixer.



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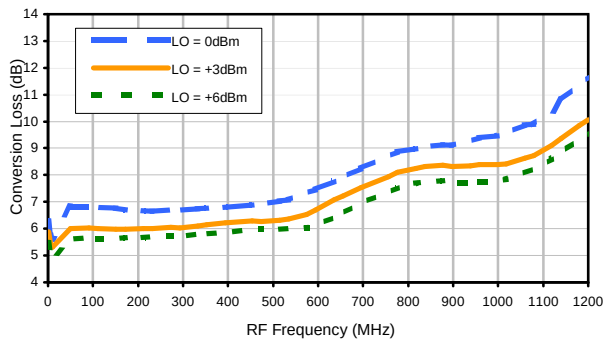
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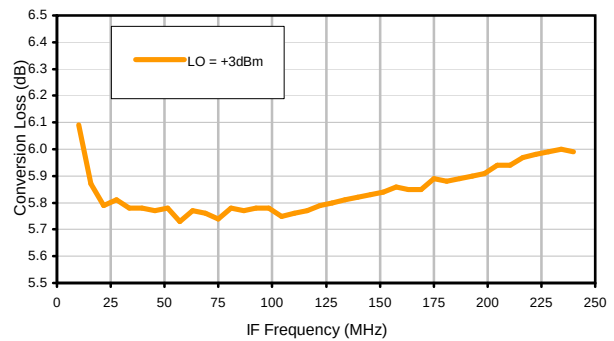
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Typical Performance Curves

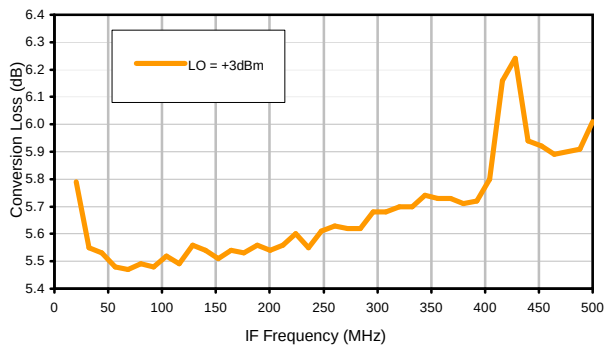
Conversion Loss @ IF=30MHz



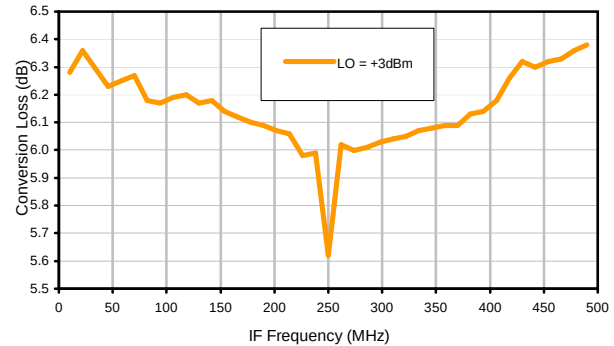
Conversion Loss vs. IF @ RF=250.1MHz



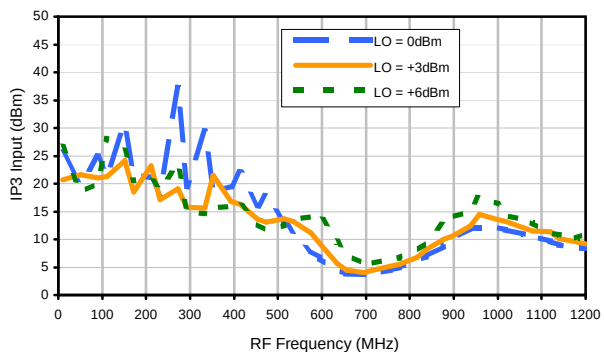
Conversion Loss vs. IF @ RF=10.1MHz



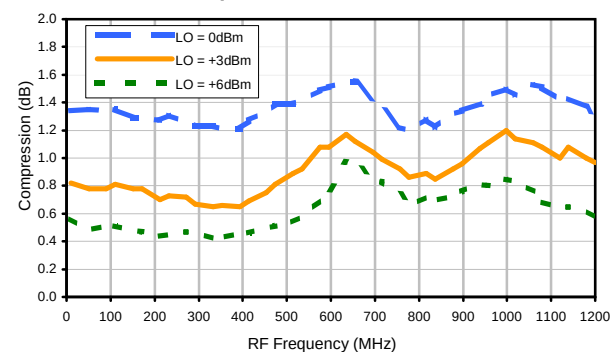
Conversion Loss vs. IF @ RF=500.1MHz



IP3 Input

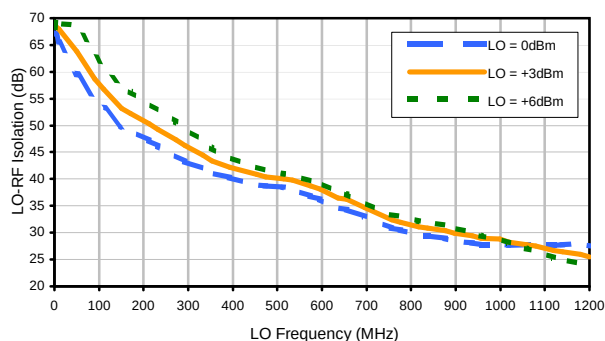


Compression @ RF IN=0dBm

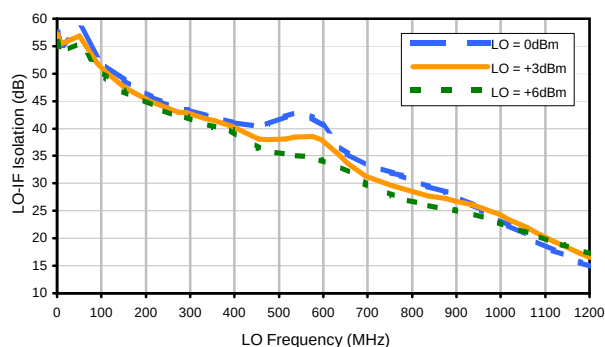


Typical Performance Curves

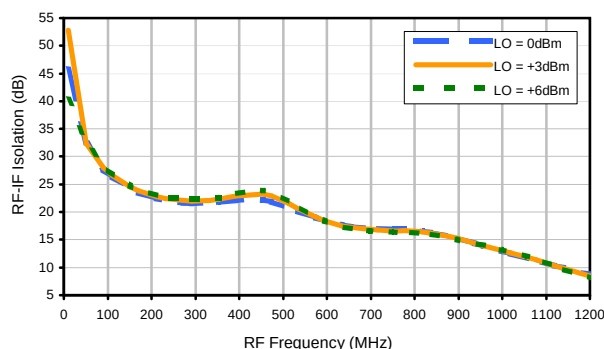
LO-RF Isolation



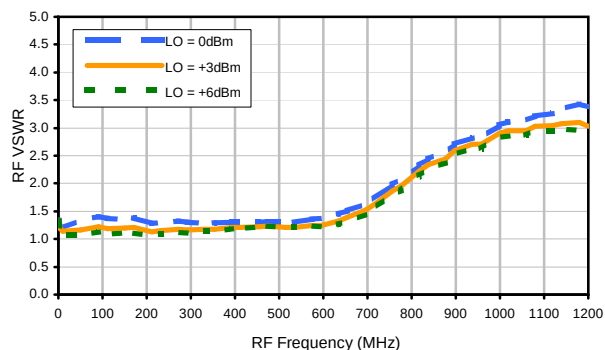
LO-IF Isolation



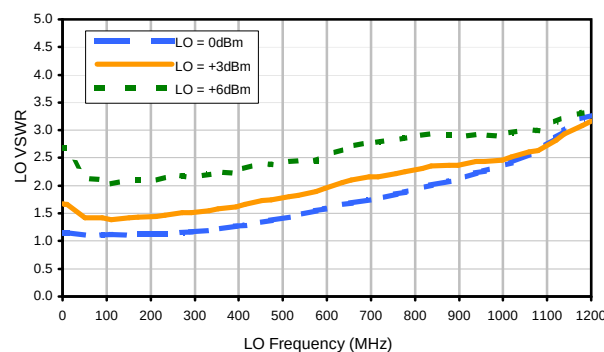
RF-IF Isolation



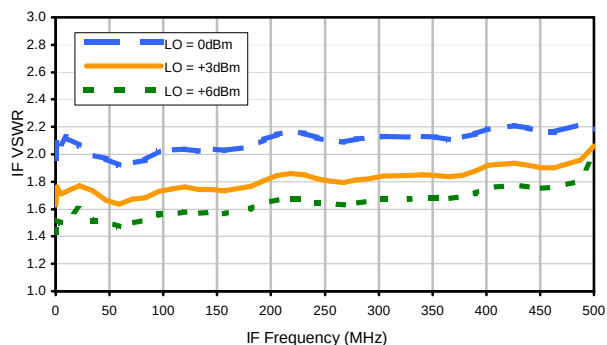
RF VSWR



LO VSWR



IF VSWR



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Harmonics Tables

RF HARMONICS ORDER

	(-dBm)	(-dBc)										
0	-	-	18	33	21	27	21	46	36	47	39	57
1	-	16	0	27	14	33	23	38	39	46	41	61
2	107	73	62	63	62	61	58	73	54	70	64	69
3	111	63	70	69	61	86	57	73	59	79	64	83
4	116	89	95	89	85	78	85	83	84	89	92	102
5	121	94	87	96	84	83	77	89	86	97	84	94
6	131	103	102	92	106	96	81	91	92	105	116	101
7	114	101	101	104	98	96	89	77	86	100	101	102
8	118	100	106	114	113	98	101	99	70	93	91	98
9	117	110	108	108	113	105	97	93	100	68	103	92
10	128	107	100	111	104	97	116	97	101	96	67	90
RF CAL	0	1	2	3	4	5	6	7	8	9	10	

LO HARMONICS ORDER

Test conditions:

RF IN: 250.1 MHz; -15.00 dBm.
LO IN: 280.01 MHz; +3.00 dBm
IF OUT: 29.91 MHz; -20.89 dBm

RF HARMONICS ORDER

	(-dBm)	(-dBc)										
0	-	-	27	42	33	40	34	54	52	72	59	70
1	-	17	0	28	14	37	24	41	43	52	52	66
2	98	58	50	58	52	63	51	57	48	72	61	67
3	93	45	47	49	50	53	48	51	52	57	55	64
4	90	68	70	69	79	66	77	65	70	71	66	80
5	93	70	67	67	58	69	54	64	53	77	56	73
6	97	83	88	81	93	96	77	87	78	82	79	81
7	93	79	77	86	73	87	74	75	70	74	69	76
8	96	95	92	89	94	88	103	93	89	92	95	92
9	97	102	89	104	81	100	81	92	89	71	96	89
10	95	103	101	103	107	100	97	90	96	94	83	95
RF CAL	0	1	2	3	4	5	6	7	8	9	10	

LO HARMONICS ORDER

Test conditions:

RF IN: 250.1 MHz; -5.00 dBm.
LO IN: 280.01 MHz; +3.00 dBm
IF OUT: 29.91 MHz; -11.01 dBm

- Notes: 1. All Harmonics are in (dBc) relative to IF OUTPUT.
2. + entry denotes harmonics are in (dBc) above IF OUTPUT.
3. RF Cal represent the Harmonics level of the RF input signal to the mixer.



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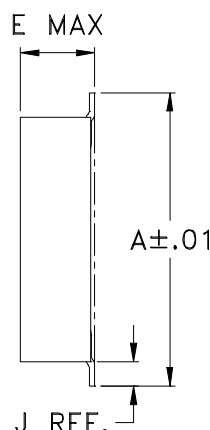
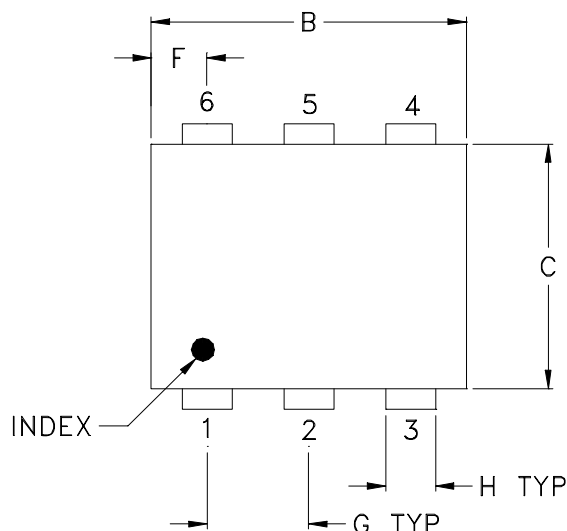
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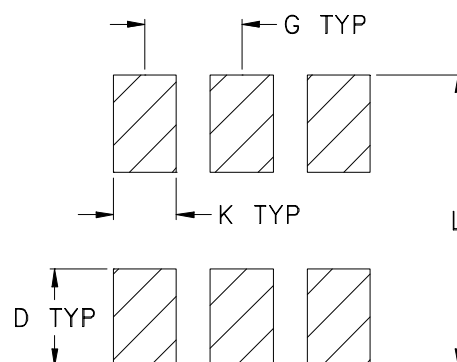
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CD541
CD542
CD636
CD637

Outline Dimensions



PCB Land Pattern



Suggested Layout,
Tolerance to be within $\pm .002$

CASE#	A	B	C	D	E	F	G	H	J	K	L	WT, GRAM
CD541					.082 (2.08)							.15
CD542	.272 (6.91)	.310 (7.87)	.220 (5.58)	.100 (2.54)	.112 (2.84)	.055 (1.40)	.100 (2.54)	.030 (0.76)	.026 (0.66)	.065 (1.65)	.300 (7.62)	.20
CD636					.162 (4.11)							.25
CD637					.206 (5.23)							.40

Dimensions are in inches (mm). Tolerances: 2 Pl. $\pm .01$; 3 Pl. $\pm .005$

Notes:

- Case material: Plastic.
- Termination finish:
For RoHS Case Styles: Tin plate over Nickel plate. All models, (+) suffix.
For RoHS-5 Case Styles: Tin-Lead plate. All models, no (+) suffix.



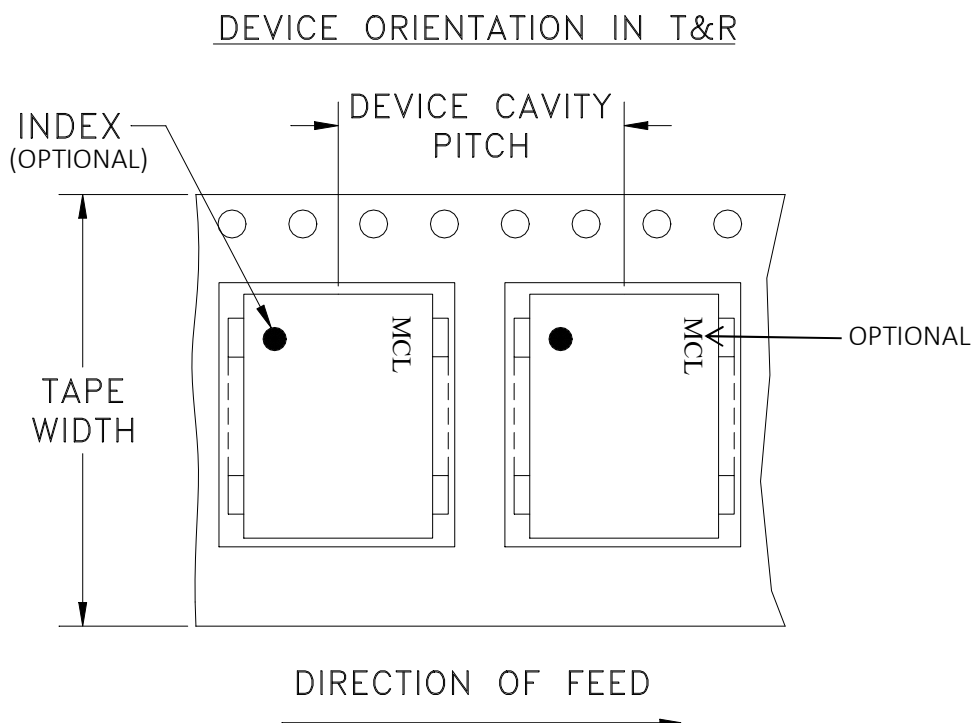
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Tape & Reel Packaging TR-F34



Tape Width, mm	Device Cavity Pitch, mm	Reel Size, inches	Devices per Reel see note	
16	12	7	Small quantity standard (see note)	20
				50
				100
				200
		13	Standard	500
				1000

Note: Availability of small reel quantity varies by model.
Refer to pricing and availability on individual model dashboard.

Mini-Circuits carrier tape materials provide protection from ESD (Electro-Static Discharge) during handling and transportation. Tapes are static dissipative and comply with industry standards EIA-481/EIA-541.

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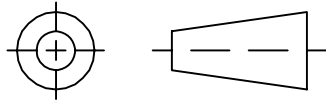
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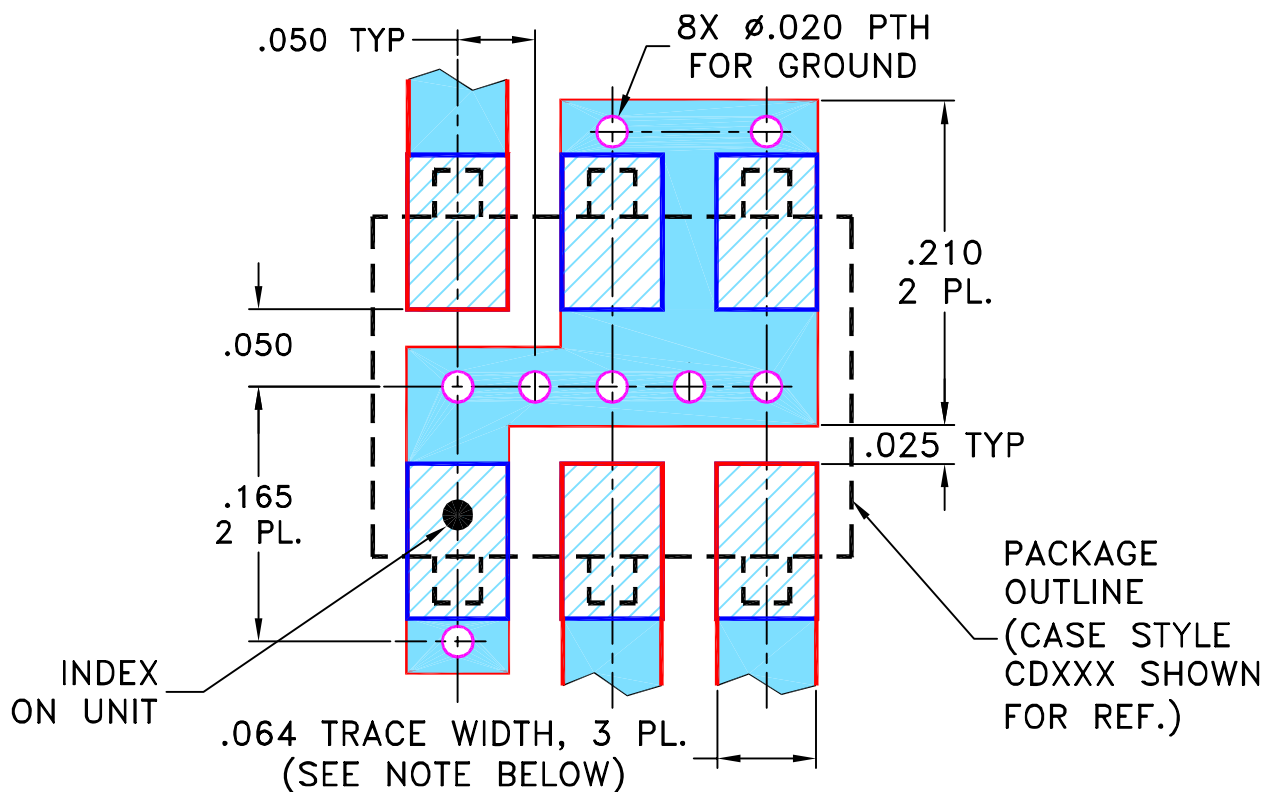
THIRD ANGLE PROJECTION



REVISIONS

REV	ECN No.	DESCRIPTION	DATE	DR	AUTH
A	M101143	ADDED "gk" PIN CONNECTION, TT100 CASE STYLE & NOTE 2	10/10/05	MMG	DJ
B	M102713	ADDED "...WITH SMOBC"	01/17/06	MMG	IL
C	M108637	REMOVED "PIN 1", ADDED INDEX ON UNIT	12/01/06	MYG	FL

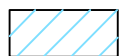
SUGGESTED MOUNTING CONFIGURATION
FOR BH292, CD541/542/636/637, TT100/240 CASE
STYLES, "gk", "ht", "hu", "nd", "w" PIN CONNECTIONS



- NOTES:** 1. TRACE WIDTH IS SHOWN FOR ROGERS R04350B WITH DIELECTRIC THICKNESS .030" $\pm$.002"; COPPER: 1/2 OZ. EACH SIDE.
 FOR OTHER MATERIALS TRACE WIDTH MAY NEED TO BE MODIFIED.
 2. BOTTOM SIDE OF THE PCB IS CONTINUOUS GROUND PLANE.



DENOTES PCB COPPER LAYOUT WITH SMOBC
 (SOLDER MASK OVER BARE COPPER)



DENOTES COPPER LAND PATTERN FREE OF SOLDER MASK

UNLESS OTHERWISE SPECIFIED

INITIALS

DATE

DIMENSIONS ARE IN INCHES

TOLERANCES ON:

2 PL DECIMALS $\pm$ 3 PL DECIMALS $\pm$.005ANGLES $\pm$ FRACTIONS $\pm$

DRAWN

MMG

07/17/02

CHECKED

WL

08/02/02

APPROVED

DJ

08/05/02



Mini-Circuits®

13 Neptune Avenue
 Brooklyn NY 11235

PL, gk/ht/hu/nd/w, BH292,
 CD541/542/636/637, TT100/240, TB-03

SIZE

A

CODE IDENT

15542

DRAWING NO:

98-PL-052

REV:

C

FILE:

98PL052

SCALE:

8:1

SHEET:

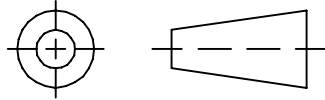
1 OF 1

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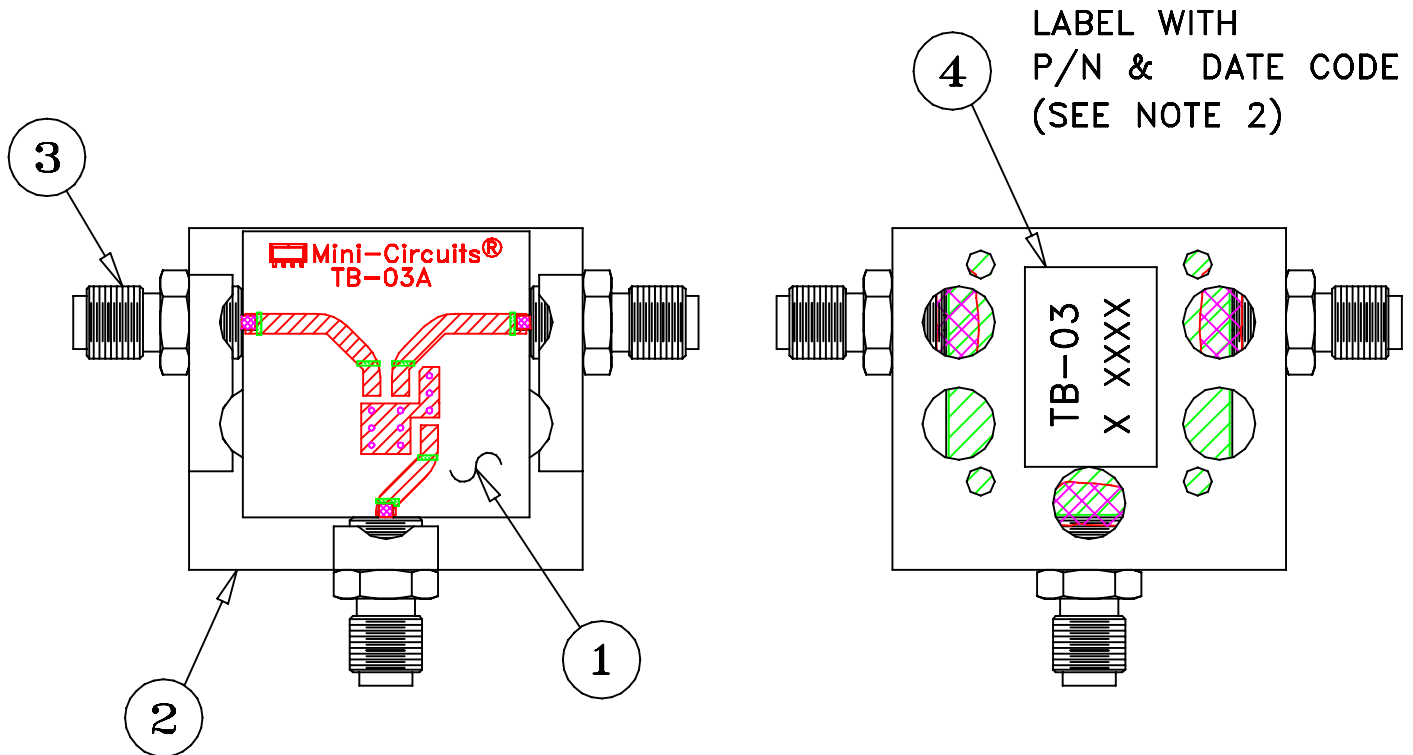
ASHEETA1.DWG REV:A DATE:01/12/95

THIRD ANGLE PROJECTION



REVISIONS

REV	ECN No.	DESCRIPTION	DATE	DR	AUTH
E	M119737	UPDATED PCB	10.08	MF	AD
F	M127659	UPDATED CARR	06.10	SW	SG
G	M127846	UPDATED SCHEMATIC DIAGRAM	06.10	SW	SG
H	M131840	UPDATED DWG	05.11	MF	AD

**NOTES:**

1. REFER TO -09 PAGE FOR ITEM DESCRIPTIONS.
DESIGNATION NUMBERS ON -20 PAGE CORRESPOND TO THE NUMBERS ON -09 PAGE.
2. FOR TEXT HEIGHT & STYLE ON THE LABEL REFER TO: D3-G209.

UNLESS OTHERWISE SPECIFIED

INITIALS

DATE

DIMENSIONS ARE IN INCHES

DRAWN

S.WOLYNSKI

08.29.99

TOLERANCES ON:

CHECKED

SG

07.06.99

2 PL DECIMALS ±

APPROVED

MG

07.10.99

3 PL DECIMALS ±

ANGLES ±

FRACTIONS ±

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ASHEETA1.DWG REV:A DATE:01/12/95


Mini-Circuits® 13 Neptune Avenue
Brooklyn NY 11235

TB,ADE,CD542/636,06MX01,50

SIZE

A

CODE IDENT

15542

DRAWING NO:

TB-03-20

REV:

H

FILE:

WTB-03

SCALE:

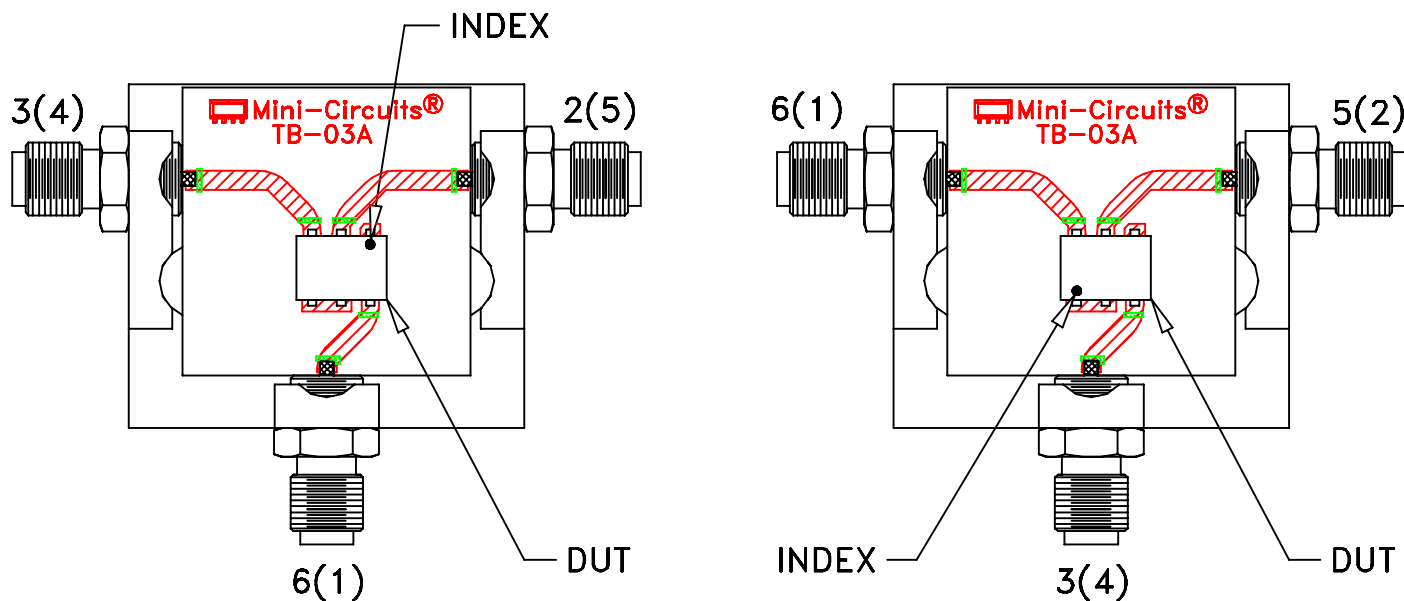
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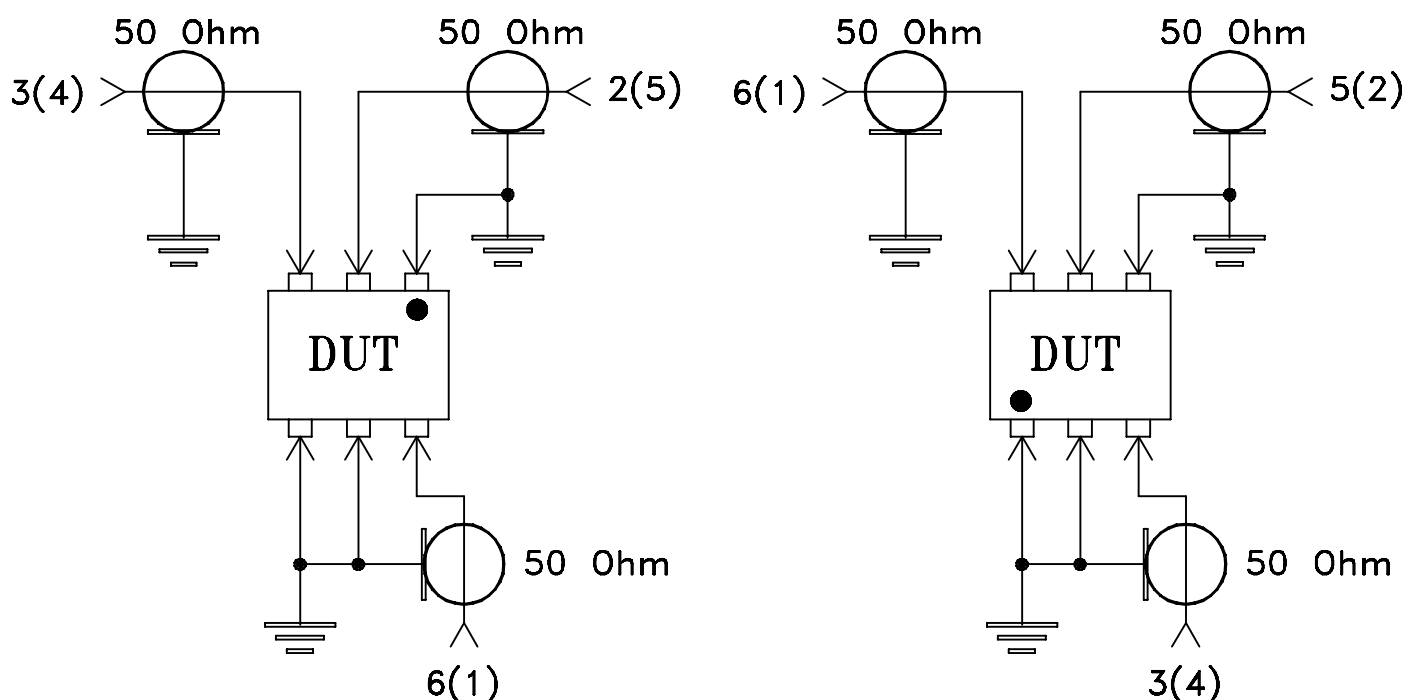
1 OF 2

Evaluation Board and Circuit

For Pin Connections and DUT Orientation Refer to
Data Sheet of the DUT



TB-03



Schematic Diagram

Notes:

1. 50 Ohm SMA Female connectors.
2. PCB Material: Rogers R04350 or equivalent,
Dielectric Constant=3.5, Thickness=.030 inch.

Mini-Circuits®



All Mini-Circuits products are manufactured under exacting quality assurance and control standards, and are capable of meeting published specifications after being subjected to any or all of the following physical and environmental test.

Specification	Test/Inspection Condition	Reference/Spec
Operating Temperature	-40° to 85°C Ambient Environment	Individual Model Data Sheet
Storage Temperature	-55° to 100° C Ambient Environment	Individual Model Data Sheet
Humidity	90 to 95% RH, 240 hours, 50°C	MIL-STD-202, Method 103, Condition A, Except 50°C and end-point electrical test done within 12 hours
Thermal Shock	-55° to 100°C, 100 cycles	MIL-STD-202, Method 107, Condition A-3, except +100°C
Solder Reflow Heat	Sn-Pb Eutetic Process: 225°C peak Pb-Free Process 245° - 250°C peak	J-STD-020, Table 4-1, 4-2 and 5-2, Figure 5-1
Solderability	10X Magnification	J-STD-002, 95% Coverage
Vibration (High Frequency)	20g peak, 10-2000 Hz, 12 times in each of three perpendicular directions (total 36)	MIL-STD-202, Method 204, Condition D
Mechanical Shock	50g, 11 ms, 1/2-sine, 18 shocks: 3 each direction, each of 3 axes	MIL-STD-202, Method 213, Condition A
Marking Resistance to Solvents	Isopropyl alcohol + mineral spirits at 25°C; terpene defluxer at 25°C; distilled water + proylene glycol monomethyl ether + monoethanolamine at 63°C to 70°C	MIL-STD-202, Method 215